



Evaluating Key Criteria for Railway Timetable Planning: A Fuzzy PIPRECIA-Based Decision Framework for Transportation System Operations

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Abstract: Railway timetable planning plays a central role in the coordination and operational performance of transportation systems. Efficient timetable development remains essential for balancing infrastructure constraints, service quality, operational efficiency, and economic objectives in railway operations. The interaction among these dimensions makes timetable planning a complex decision problem for infrastructure managers and transport operators. This study aims to evaluate the relative importance of the principal criteria influencing railway timetable planning and to provide quantitative support for transportation system decision-making. A structured evaluation framework was developed using the fuzzy Pivot Pairwise Relative Criteria Importance Assessment (fuzzy PIPRECIA) method. Forty decision-makers with professional experience in railway operation, infrastructure management, engineering practice, and academia participated in the assessment process. Five main criteria were examined: railway line capacity, railway station capacity, number of passed trains, quality of train operations, and revenues of the planned timetable. The results showed that revenues of the planned timetable received the highest importance weight, followed by quality of train operations, number of passed trains, railway line capacity, and railway station capacity. The findings further showed that operational and economic dimensions exerted greater influence on timetable planning decisions than infrastructure-capacity factors. The results indicate that railway timetable planning should be approached as a system-level coordination problem rather than a capacity allocation exercise alone. This study provides a structured decision-support perspective for evaluating competing planning priorities and offers a practical basis for improving timetable development and operational performance in railway transportation systems.

Keywords: Railway timetable planning; Intelligent transportation systems; Fuzzy PIPRECIA; Transportation decision support; Railway operations; System evaluation

1 Introduction

Railway timetable planning represents one of the fundamental operational tasks in railway transportation systems. Timetables determine how infrastructure resources, train movements, operational priorities, and service requirements are coordinated within a constrained transportation environment. Effective timetable development directly affects transport capacity, operational reliability, service quality, infrastructure utilization, and economic performance. As railway systems continue to operate under increasing pressure from growing mobility demand, limited infrastructure expansion, and higher expectations regarding efficiency and service continuity, timetable planning remains an important mechanism for maintaining stable and responsive transport operations.

The development and implementation of railway timetables involve balancing multiple and often conflicting objectives. Infrastructure managers seek efficient use of line and station capacity while maintaining operational stability and safety, whereas transport operators aim to maximize transport output and economic returns under limited operational costs. These objectives are further influenced by constraints related to train routing, infrastructure

availability, operational quality, and transport demand. Consequently, timetable planning cannot be treated as an isolated scheduling activity but rather as a system-level decision process in which infrastructure conditions, operational performance, and economic outcomes interact simultaneously.

In such environments, decision-making requires structured and reliable evaluation approaches capable of incorporating multiple criteria and reducing uncertainty. Previous experience suggests that increasing system complexity makes it difficult for decision-makers to recognize and quantify the relative influence of competing planning dimensions. Decision-making therefore requires clearly defined objectives and supporting analytical procedures capable of integrating technical, operational, economic, and organizational considerations. Multi-criteria decision-making (MCDM) methods have become widely adopted because they provide mechanisms for comparing alternatives and determining the relative significance of evaluation dimensions. The selection of criteria and determination of their weights play an important role in supporting strategic and operational decisions [1–3].

Among existing approaches, fuzzy Pivot Pairwise Relative Criteria Importance Assessment (fuzzy PIPRECIA) has demonstrated applicability in situations characterized by uncertainty and subjective judgement. Vesković et al. [4] applied fuzzy PIPRECIA to determine the significance of criteria in the selection of container handling equipment and established a structured framework consisting of multiple influencing groups. Pamučar and Čirović [5] employed Multi-Attributive Border Approximation area Comparison (MABAC) for selecting transport and manual resources in logistics centres. These studies demonstrate the usefulness of weighting methods in supporting transport-related decisions where multiple objectives must be evaluated simultaneously.

Within railway transportation research, numerous studies have adopted multi-criteria approaches for evaluating operational and infrastructure-related problems. Vesković et al. [6] proposed an integrated Delphi–SWARA–MABAC framework for assessing railway management restructuring. Krmac and Djordjević [7] used the Analytic Hierarchy Process method to evaluate performance indicators for intelligent transportation systems, while Aydin [8] investigated railway line performance using technique for order preference by similarity to ideal solution. Leonardi [9] applied Analytic Hierarchy Process to railway infrastructure planning, and Santarremigia et al. [10] used multi-criteria assessment to support railway transport safety decisions involving hazardous materials. Stević et al. [11] determined the importance of wagon selection criteria using BWM and SAW methods. In parallel, several studies have addressed timetable planning problems and timetable evaluation under different operational settings [12–16].

Although previous studies have demonstrated the usefulness of MCDM approaches in railway applications, limited attention has been devoted to systematically identifying and quantifying the relative importance of the principal dimensions that govern railway timetable planning itself. Existing studies frequently focus on infrastructure design, transport performance, safety assessment, or evaluation of timetable alternatives, while the prioritization of planning criteria before timetable implementation remains insufficiently explored. In addition, interactions among operational quality, transport throughput, infrastructure constraints, and economic outcomes often remain implicitly addressed.

Therefore, this study aims to evaluate the relative importance of the main criteria influencing railway timetable planning and implementation and to determine their quantitative significance through a structured decision-support framework. Five principal criteria were defined based on long-term professional experience: railway line capacity, railway station capacity, number of passed trains, quality of train operations, and revenues of the planned timetable. To quantify their relative importance, the fuzzy PIPRECIA method was applied using evaluations collected from forty decision-makers with professional experience in railway operation, infrastructure management, engineering practice, and academia.

The remainder of the paper is organized as follows. Section 2 describes the criteria influencing railway timetable planning. Section 3 presents the fuzzy PIPRECIA methodology. Section 4 reports and discusses the obtained results. Finally, Section 5 summarizes the findings, limitations, and future research directions.

2 Brief Description of the Criteria Influencing Railway Timetable Planning

Railway timetable planning is determined by the interaction of infrastructure conditions, operational constraints, transport demand, and economic objectives. In practical railway operations, timetable decisions are not governed by a single factor but emerge from balancing multiple dimensions that jointly influence transportation system performance. Based on professional experience and expert knowledge, five principal criteria were selected and evaluated in this study:

- C1: Railway line capacity
- C2: Railway station capacity
- C3: Number of passed trains
- C4: Quality of train operations
- C5: Revenues of the planned timetable

Together, these criteria represent infrastructure capability, operational throughput, service performance, and economic outcomes within railway transportation systems.

C1: Railway line capacity

Railway line capacity reflects the operational ability of infrastructure segments managed by the infrastructure manager to accommodate train movements under specified operating conditions. This criterion determines the upper limit of transport throughput and influences timetable feasibility, operational flexibility, and service reliability. The following elements are considered:

- 1) Number of tracks on the railway line: directly affects the number of trains that can operate simultaneously and contributes to transport capacity;
- 2) Permitted speed on the railway line: affects running time, travel time, and line utilization efficiency;
- 3) Mode of train operations: influences capacity and operational quality through train control and signalling arrangements (classical operation, inter-station dependency, block spacing, remote control, ETCS);
- 4) Signal position on the line: determines train separation and affects traffic density along the line;
- 5) Reduced speed on the line: decreases effective line capacity and limits timetable flexibility.

C2: Railway station capacity

Railway station capacity represents the ability of station infrastructure to receive, process, and dispatch train services while maintaining operational continuity. Station performance affects timetable robustness and train circulation efficiency. This criterion includes:

- 1) Number of tracks within the station: influences simultaneous train accommodation capability;
- 2) Permitted speed within the station: affects movement efficiency over tracks and switches;
- 3) Station security (signal position and signal type): determines operational intervals and traffic organization;
- 4) Track scheme and route interdependence scheme: describes route characteristics, route interaction, and through-running capability;
- 5) Reduced speed within the station: negatively affects operational throughput and timetable execution.

C3: Number of passed trains

The number of passed trains represents transport throughput and indicates the operational output achievable within a defined period. This criterion reflects the practical implementation of timetable planning under infrastructure and operational constraints. The following elements are considered:

- 1) Type and number of trains on a railway line: represent transport demand and infrastructure utilization;
- 2) Number of introduced and cancelled trains: reflects transport requirements and operational feasibility;
- 3) Travel time: depends on infrastructure conditions and operational requests;
- 4) Number of transported passengers / mass of transported goods: reflects transport effectiveness and service provision;
- 5) Average train speed: represents operational efficiency of transport services.

C4: Quality of train operations

Quality of train operations reflects the extent to which actual train services conform to planned timetable conditions. This criterion characterizes timetable stability, service reliability, and operational performance. The following elements are included:

- 1) Planned transport times: indicate alignment between timetable design and transport demand;
- 2) Reserve running and travel times: influence timetable stability and implementation quality;
- 3) Unplanned train stops: represent disturbances affecting operational continuity;
- 4) Connection availability at intermediate stations: reflects passenger transfer capability and network connectivity;
- 5) Time intervals for infrastructure maintenance: represent operational windows allocated for maintaining railway infrastructure.

C5: Revenues of the planned timetable

Revenues of the planned timetable represent the economic outcomes generated by timetable implementation while considering associated operational and infrastructure costs. This criterion captures the financial consequences of planning decisions and includes:

- 1) Revenues from planned train routes: represent expected income generated under scheduled operations;
- 2) Revenues from subsequently planned “ad hoc” train routes: indicate additional income generated through timetable adjustments;
- 3) Revenues from train operations: represent realized operational income under regular and ad hoc services;
- 4) Costs of carriers due to train stops: indicate economic losses caused by operational disruptions;
- 5) Infrastructure costs due to train stops: represent expenditures associated with planned and unplanned interruptions.

These five criteria collectively describe the principal dimensions governing railway timetable planning and provide the basis for subsequent quantitative evaluation using the fuzzy PIPRECIA framework.

3 Fuzzy Pivot Pairwise Relative Criteria Importance Assessment

The PIPRECIA method was developed in study [17]. The fuzzy PIPRECIA method consists of 11 steps, which are presented below, as described by Stević et al. [1] and Đalić et al. [18]. This method has proven its applicability through its exploitation [19–25].

Step 1. Formation of a required set of criteria, labeling of criteria, formation of groups and subgroups of criteria and establishment of a group of experts for evaluating the significance of the criteria.

Step 2. Experts' evaluation of the defined set of criteria using the scales provided in Table 1, starting from the second to the last criterion (from C_2 to C_n), according to Eq. (1).

$$\bar{s}_j^r = \begin{cases} > \bar{1} & \text{if } C_j > C_{j-1} \\ = \bar{1} & \text{if } C_j = C_{j-1} \\ < \bar{1} & \text{if } C_j < C_{j-1} \end{cases} \quad (1)$$

Table 1. Scale 1–2 for the evaluation of criteria (PIPRECIA)

Description of Value	Score	Scale	l	m	u	DFV
Nearly equal significance	1+	—	1.000	1.000	1.050	1.008
Slightly more significant	2	—	1.100	1.150	1.200	1.150
Moderately more significant	3	—	1.200	1.300	1.350	1.292
More significant	4	1–2	1.300	1.450	1.500	1.433
Much more significant	5	—	1.400	1.600	1.650	1.575
Dominantly more significant	6	—	1.500	1.750	1.800	1.717
Absolutely more significant	7	—	1.600	1.900	1.950	1.858

Note: PIPRECIA = Pivot Pairwise Relative Criteria Importance Assessment; DFV = defuzzified value. l , m , and u denote the lower, modal, and upper values of the triangular fuzzy number, respectively.

The evaluation of the criteria by the decision-maker r is denoted by the symbol $\bar{s}_j^r$. To obtain the matrix $\bar{s}_j$, it is necessary to average the matrix $\bar{s}_j^r$ using the geometric mean, arithmetic mean or by applying a specific aggregator.

When the criterion being evaluated is of greater significance than the previous criterion, the evaluation is conducted using the scale in Table 1, and when it is of lesser significance than the previous criterion, using the scale in Table 2. Tables 1 and 2 show the defuzzified value (DFV) for each comparison in order to allow for easier evaluation of the criteria.

Table 2. Scale 0–1 for the evaluation of criteria (PIPRECIA-I)

l	m	u	DFV	Scale	Score	Description of Value
0.667	1.000	1.000	0.944	—	1-	Slightly less significant
0.500	0.667	1.000	0.694	—	1/2	Moderately less significant
0.400	0.500	0.667	0.511	—	1/3	Less significant
0.333	0.400	0.500	0.406	0–1	1/4	Very less significant
0.286	0.333	0.400	0.337	—	1/5	Much less significant
0.250	0.286	0.333	0.288	—	1/6	Dominantly less significant
0.222	0.250	0.286	0.251	—	1/7	Absolutely less significant

Note: PIPRECIA = Pivot Pairwise Relative Criteria Importance Assessment; DFV = defuzzified value. l , m , and u denote the lower, modal, and upper values of the triangular fuzzy number, respectively.

Step 3. Determination of the coefficient $\bar{k}_j$,

$$\bar{k}_j = \begin{cases} = \bar{1} & \text{if } j = 1 \\ 2 - \bar{s}_j & \text{if } j > 1 \end{cases} \quad (2)$$

Step 4. Determination of the fuzzy weight $\bar{q}_j$,

$$\bar{q}_j = \begin{cases} = \bar{1} & \text{if } j = 1 \\ \frac{\bar{q}_{j-1}}{\bar{k}_j} & \text{if } j > 1 \end{cases} \quad (3)$$

Step 5. Determination of the relative weight of the criterion $\bar{w}_j$.

$$\bar{w}_j = \frac{\bar{q}_j}{\sum_{j=1}^n \bar{q}_j} \quad (4)$$

In the following section, the methodology for the inverse fuzzy PIPRECIA method is applied.

Step 6: The criteria defined by the scale in Table 2 are evaluated starting from the penultimate to the first criterion (from C_{n-1} to C_1), according to Eq. (5).

$$\bar{s}_j^{r'} = \begin{cases} > \bar{1} & \text{if } C_j > C_{j+1} \\ = \bar{1} & \text{if } C_j = C_{j+1} \\ < \bar{1} & \text{if } C_j < C_{j+1} \end{cases} \quad (5)$$

The evaluation of the criteria by the decision-maker r is denoted by the symbol $\bar{s}_j^{r'}$. In order to obtain the matrix $\bar{s}_j'$, it is necessary to average the matrix $\bar{s}_j^{r'}$ using the geometric mean.

Step 7. Determination of the coefficient $\bar{k}_j'$.

$$\bar{k}_j' = \begin{cases} = \bar{1} & \text{if } j = n \\ 2 - \bar{s}_j' & \text{if } j < n \end{cases} \quad (6)$$

Step 8. Determination of the fuzzy weight $\bar{q}_j'$.

$$\bar{q}_j' = \begin{cases} = \bar{1} & \text{if } j = n \\ \bar{q}_{j+1}' & \text{if } j < n \end{cases} \quad (7)$$

Step 9. Determination of the relative weight of the criterion $\bar{w}_j'$.

$$\bar{w}_j' = \frac{\bar{q}_j'}{\sum_{j=1}^n \bar{q}_j'} \quad (8)$$

Step 10. To determine the final weights of the criteria, it is necessary to defuzzify the fuzzy values $\bar{w}_j$ and $\bar{w}_j'$.

$$\bar{w}_j'' = \frac{1}{2} (\bar{w}_j + \bar{w}_j') \quad (9)$$

Step 11. Verification of the results obtained by applying the Spearman and Pearson correlation coefficients.

4 Evaluating Key Criteria for Railway Timetable Planning Using Fuzzy Pivot Pairwise RElative Criteria Importance Assessment

To evaluate the relative importance of the dimensions governing railway timetable planning, five principal criteria identified in Section 2 were assessed using the fuzzy PIPRECIA framework: railway line capacity (C1), railway station capacity (C2), number of passed trains (C3), quality of train operations (C4), and revenues of the planned timetable (C5). These criteria collectively represent infrastructure capability, operational performance, transport throughput, and economic outcomes within railway transportation systems.

The evaluation process was conducted through expert judgement involving a total of 40 decision-makers. In order to ensure that the assessment reflected practical and strategic perspectives relevant to timetable planning, the expert profile was examined from four aspects: occupation, work experience, qualification level, and employer category (Figure 1).

Regarding occupation structure, three professional groups were represented. The largest proportion of experts consisted of managers (24), followed by design engineers (10) and professors (6). In terms of professional experience, the majority of respondents had between 20 and 30 years of work experience (18), while 11 experts had between 30 and 40 years, 10 experts had between 10 and 20 years, and one expert had less than 10 years of experience. This distribution ensured the inclusion of long-term operational and planning perspectives.

The qualification profile included three levels. Twenty-nine experts belonged to qualification level VII, eight experts to qualification level VIII, and three experts to qualification level VI. With respect to institutional affiliation, most participants were employed by railway infrastructure organizations (18), followed by transport operators (11), academic institutions (6), and public administration bodies (5). The composition of the expert panel was intended to provide balanced representation of infrastructure, operation, managerial, and analytical viewpoints relevant to railway timetable development.

The evaluation of criteria was performed using the linguistic assessment scales presented in Table 1 and Table 2, which were subsequently transformed into fuzzy triangular numbers for quantitative analysis. Table 3 presents the evaluation results obtained using the fuzzy PIPRECIA method, whereas Table 4 reports the corresponding results obtained through the fuzzy PIPRECIA-Inverse procedure. The integration of both evaluation directions enables a more robust estimation of criterion significance and supports a structured interpretation of planning priorities within railway transportation systems.

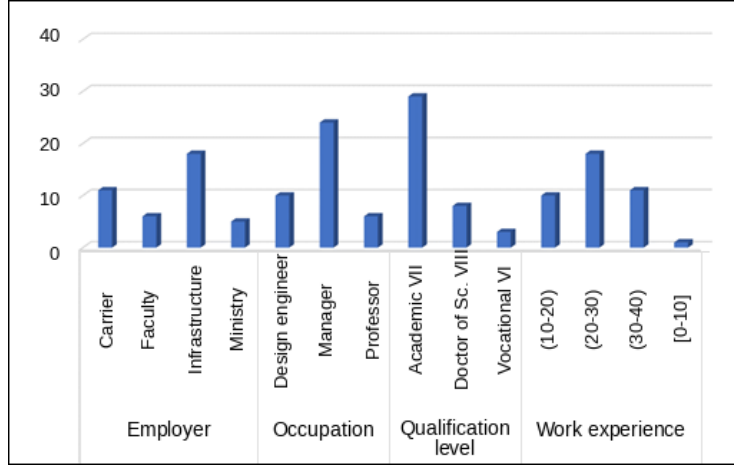


Figure 1. Profile of experts

Based on the evaluation of the criteria and applying Eq. (2), the matrix $\bar{s}_j$ is formed.

$$\bar{s}_j = \begin{bmatrix} 0.709 & 0.847 & 0.962 \\ 1.112 & 1.275 & 1.327 \\ 0.898 & 1.012 & 1.122 \\ 0.843 & 0.952 & 1.059 \end{bmatrix}$$

Using Eq. (3), the matrix $\bar{k}_j$ is formed.

$$\bar{k}_j = \begin{bmatrix} 1.000 & 1.000 & 1.000 \\ 1.038 & 1.153 & 1.291 \\ 0.673 & 0.725 & 0.888 \\ 0.878 & 0.898 & 1.102 \\ 0.941 & 1.048 & 1.157 \end{bmatrix}$$

Using Eq. (4), the matrix $\bar{q}_j$ is formed.

$$\bar{q}_j = \begin{bmatrix} 1.000 & 1.000 & 1.000 \\ 0.775 & 0.867 & 0.963 \\ 0.872 & 1.197 & 1.430 \\ 0.792 & 1.211 & 1.629 \\ 0.684 & 1.156 & 1.731 \end{bmatrix}$$

Before forming the matrix of relative weights of the criterion $\bar{w}_j$, as an intermediate step, the values for $\bar{q}_j$ are summed up and the values obtained are: (4.123; 5.431; 6.754).

By applying Eq. (5), the matrix of relative weights of the criterion $\bar{w}_j$ is formed.

$$\bar{w}_j = \begin{bmatrix} 0.148 & 0.184 & 0.243 \\ 0.115 & 0.160 & 0.234 \\ 0.129 & 0.220 & 0.347 \\ 0.117 & 0.223 & 0.395 \\ 0.101 & 0.213 & 0.420 \end{bmatrix}$$

After applying the equation: $DF = \frac{l+4m+u}{6}$, the defuzzy-I values obtained for the PIPRECIA method are: 0.188; 0.164; 0.226; 0.234 and 0.229.

In order to determine the final weights of the criteria, it is necessary to apply the equations for the inverse fuzzy PIPRECIA method.

Based on the evaluation of the criteria and applying Eq. (6), the matrix $\bar{s}'_j$ is formed.

$$\bar{s}'_j = \begin{bmatrix} 0.919 & 1.035 & 1.115 \\ 0.525 & 0.601 & 0.738 \\ 0.744 & 0.881 & 0.947 \\ 0.794 & 0.932 & 1.003 \end{bmatrix}$$

By applying Eq. (7), the matrix $\bar{k}'_j$ is formed.

$$\bar{k}'_j = \begin{bmatrix} 0.885 & 0.965 & 1.081 \\ 1.262 & 1.399 & 1.475 \\ 1.053 & 1.119 & 1.256 \\ 0.997 & 1.068 & 1.206 \\ 1.000 & 1.000 & 1.000 \end{bmatrix}$$

Table 3. Ratings of the main criteria using the fuzzy PIPRECIA method

No.	C2			C3			C4			C5		
	<i>k</i>	<i>m</i>	<i>u</i>	<i>k</i>	<i>m</i>	<i>u</i>	<i>k</i>	<i>m</i>	<i>u</i>	<i>k</i>	<i>m</i>	<i>u</i>
1	0.400	0.500	0.667	1.300	1.450	1.500	1.200	1.300	1.350	0.286	0.333	0.400
2	0.667	1.000	1.000	1.400	1.600	1.650	1.200	1.300	1.350	1.100	1.150	1.200
3	0.667	1.000	1.000	0.400	0.500	0.667	1.100	1.150	1.200	0.333	0.400	0.500
4	0.333	0.400	0.500	1.400	1.600	1.650	1.200	1.300	1.350	0.400	0.500	0.667
5	0.333	0.400	0.500	1.300	1.450	1.500	0.333	0.400	0.500	1.300	1.450	1.500
6	0.400	0.500	0.667	1.100	1.150	1.200	0.400	0.500	0.667	1.300	1.450	1.500
7	0.500	0.667	1.000	1.300	1.450	1.500	0.400	0.500	0.667	0.500	0.667	1.000
8	1.100	1.150	1.200	0.667	1.000	1.000	0.667	1.000	1.000	1.000	1.000	1.050
9	0.500	0.667	1.000	0.667	1.000	1.000	1.400	1.600	1.650	0.500	0.667	1.000
10	0.500	0.667	1.000	1.300	1.450	1.500	0.400	0.500	0.667	1.200	1.300	1.350
11	1.000	1.000	1.050	1.200	1.300	1.350	0.500	0.667	1.000	1.300	1.450	1.500
12	1.100	1.150	1.200	1.300	1.450	1.500	1.000	1.000	1.050	0.500	0.667	1.000
13	1.100	1.150	1.200	1.200	1.300	1.350	0.500	0.667	1.000	0.500	0.667	1.000
14	1.200	1.300	1.350	1.100	1.150	1.200	0.333	0.400	0.500	1.200	1.300	1.350
15	0.667	1.000	1.000	1.300	1.450	1.500	0.400	0.500	0.667	1.300	1.450	1.500
16	1.000	1.000	1.050	0.500	0.667	1.000	1.000	1.000	1.050	1.000	1.000	1.050
17	1.100	1.150	1.200	1.100	1.150	1.200	1.000	1.000	1.050	0.333	0.400	0.500
18	0.667	1.000	1.000	1.100	1.150	1.200	0.400	0.500	0.667	0.333	0.400	0.500
19	0.500	0.667	1.000	1.200	1.300	1.350	0.400	0.500	0.667	1.000	1.000	1.050
20	0.500	0.667	1.000	1.300	1.450	1.500	1.300	1.450	1.500	0.400	0.500	0.667
21	0.400	0.500	0.667	1.300	1.450	1.500	1.100	1.150	1.200	0.333	0.400	0.500
22	0.667	1.000	1.000	0.667	1.000	1.000	1.000	1.000	1.050	1.000	1.000	1.050
23	1.000	1.000	1.050	0.667	1.000	1.000	1.000	1.000	1.050	1.000	1.000	1.050
24	0.667	1.000	1.000	0.667	1.000	1.000	1.000	1.000	1.050	1.000	1.000	1.050
25	0.667	1.000	1.000	0.667	1.000	1.000	1.000	1.000	1.050	0.667	1.000	1.000
26	0.400	0.500	0.667	1.200	1.300	1.350	0.500	0.667	1.000	1.400	1.600	1.650
27	0.333	0.400	0.500	1.200	1.300	1.350	0.500	0.667	1.000	1.300	1.450	1.500
28	0.500	0.667	1.000	1.100	1.150	1.200	1.000	1.000	1.050	0.667	1.000	1.000
29	0.400	0.500	0.667	1.100	1.150	1.200	1.000	1.000	1.050	0.500	0.667	1.000
30	0.400	0.500	0.667	1.200	1.300	1.350	0.400	0.500	0.667	1.200	1.300	1.350
31	0.667	1.000	1.000	1.300	1.450	1.500	1.400	1.600	1.650	0.333	0.400	0.500
32	1.300	1.450	1.500	1.600	1.900	1.950	1.500	1.750	1.800	1.400	1.600	1.650
33	0.333	0.400	0.500	1.400	1.600	1.650	1.500	1.750	1.800	0.400	0.500	0.667
34	1.200	1.300	1.350	1.400	1.600	1.650	1.500	1.750	1.800	0.333	0.400	0.500
35	1.000	1.000	1.050	1.200	1.300	1.350	1.300	1.450	1.500	1.000	1.000	1.050
36	1.000	1.000	1.050	0.667	1.000	1.000	0.500	0.667	1.000	1.100	1.150	1.200
37	1.000	1.000	1.050	1.300	1.450	1.500	1.400	1.600	1.650	1.000	1.000	1.050
38	0.500	0.667	1.000	1.100	1.150	1.200	0.400	0.500	0.667	1.400	1.600	1.650
39	1.300	1.450	1.500	1.400	1.600	1.650	1.500	1.750	1.800	1.400	1.600	1.650
40	0.400	0.500	0.667	1.200	1.300	1.350	1.300	1.450	1.500	0.500	0.667	1.000

Note: PIPRECIA = Pivot Pairwise Relative Criteria Importance Assessment; C2: Railway station capacity; C3: Number of passed trains; C4: Quality of train operations; C5: Revenues of the planned timetable. l, m, and u denote the lower, modal, and upper values of the triangular fuzzy number, respectively.

By applying Eq. (8), the matrix $\bar{q}'_j$ is formed.

$$\bar{q}'_j = \begin{bmatrix} 0.414 & 0.620 & 0.852 \\ 0.447 & 0.598 & 0.754 \\ 0.660 & 0.836 & 0.952 \\ 0.829 & 0.936 & 1.003 \\ 1.000 & 1.000 & 1.000 \end{bmatrix}$$

Table 4. Ratings of the main criteria using the fuzzy PIPRECIA-Inverse method

No.	C1			C2			C3			C4		
	<i>k</i>	<i>m</i>	<i>u</i>	<i>k</i>	<i>m</i>	<i>u</i>	<i>l</i>	<i>m</i>	<i>u</i>	<i>k</i>	<i>m</i>	<i>u</i>
1	1.200	1.300	1.350	0.333	0.400	0.500	0.400	0.500	0.667	1.400	1.600	1.650
2	1.000	1.000	1.050	0.286	0.333	0.400	0.400	0.500	0.667	0.500	0.667	1.000
3	1.000	1.000	1.050	1.200	1.300	1.350	0.500	0.667	1.000	1.300	1.450	1.500
4	1.300	1.450	1.500	0.286	0.333	0.400	0.400	0.500	0.667	1.200	1.300	1.350
5	1.300	1.450	1.500	0.333	0.400	0.500	1.300	1.450	1.500	0.333	0.400	0.500
6	1.200	1.300	1.350	0.500	0.667	1.000	1.200	1.300	1.350	0.333	0.400	0.500
7	1.100	1.150	1.200	0.333	0.400	0.500	1.200	1.300	1.350	1.100	1.150	1.200
8	0.500	0.667	1.000	1.000	1.000	1.050	1.000	1.000	1.050	0.667	1.000	1.000
9	1.100	1.150	1.200	1.000	1.000	1.050	0.286	0.333	0.400	1.100	1.150	1.200
10	1.100	1.150	1.200	0.333	0.400	0.500	1.200	1.300	1.350	0.400	0.500	0.667
11	0.667	1.000	1.000	0.400	0.500	0.667	1.100	1.150	1.200	0.333	0.400	0.500
12	0.500	0.667	1.000	0.333	0.400	0.500	0.667	1.000	1.000	1.100	1.150	1.200
13	0.500	0.667	1.000	0.400	0.500	0.667	1.100	1.150	1.200	1.100	1.150	1.200
14	0.400	0.500	0.667	0.500	0.667	1.000	1.300	1.450	1.500	0.400	0.500	0.667
15	1.000	1.000	1.050	0.333	0.400	0.500	1.200	1.300	1.350	0.333	0.400	0.500
16	0.667	1.000	1.000	1.100	1.150	1.200	0.667	1.000	1.000	0.667	1.000	1.000
17	0.500	0.667	1.000	0.500	0.667	1.000	0.667	1.000	1.000	1.300	1.450	1.500
18	1.000	1.000	1.050	0.500	0.667	1.000	1.200	1.300	1.350	1.300	1.450	1.500
19	1.100	1.150	1.200	0.400	0.500	0.667	1.200	1.300	1.350	0.667	1.000	1.000
20	1.100	1.150	1.200	0.333	0.400	0.500	0.333	0.400	0.500	1.200	1.300	1.350
21	1.200	1.300	1.350	0.333	0.400	0.500	0.500	0.667	1.000	1.300	1.450	1.500
22	1.000	1.000	1.050	1.000	1.000	1.050	0.667	1.000	1.000	0.667	1.000	1.000
23	0.667	1.000	1.000	1.000	1.000	1.050	0.667	1.000	1.000	0.667	1.000	1.000
24	1.000	1.000	1.050	1.000	1.000	1.050	0.667	1.000	1.000	0.667	1.000	1.000
25	1.000	1.000	1.050	1.000	1.000	1.050	0.667	1.000	1.000	1.000	1.000	1.050
26	1.200	1.300	1.350	0.400	0.500	0.667	1.100	1.150	1.200	0.286	0.333	0.400
27	1.300	1.450	1.500	0.400	0.500	0.667	1.100	1.150	1.200	0.333	0.400	0.500
28	1.100	1.150	1.200	0.500	0.667	1.000	0.667	1.000	1.000	1.000	1.000	1.050
29	1.200	1.300	1.350	0.500	0.667	1.000	0.667	1.000	1.000	1.100	1.150	1.200
30	1.200	1.300	1.350	0.400	0.500	0.667	1.200	1.300	1.350	0.400	0.500	0.667
31	1.000	1.000	1.050	0.333	0.400	0.500	0.286	0.333	0.400	1.300	1.450	1.500
32	0.333	0.400	0.500	0.222	0.250	0.286	0.250	0.286	0.333	0.286	0.333	0.400
33	1.300	1.450	1.500	0.286	0.333	0.400	0.250	0.286	0.333	1.200	1.300	1.350
34	0.400	0.500	0.667	0.286	0.333	0.400	0.250	0.286	0.333	1.300	1.450	1.500
35	0.667	1.000	1.000	0.400	0.500	0.667	0.333	0.400	0.500	0.667	1.000	1.000
36	0.667	1.000	1.000	1.000	1.000	1.050	1.100	1.150	1.200	0.500	0.667	1.000
37	0.667	1.000	1.000	0.333	0.400	0.500	0.286	0.333	0.400	0.667	1.000	1.000
38	1.100	1.150	1.200	0.500	0.667	1.000	1.200	1.300	1.350	0.286	0.333	0.400
39	0.333	0.400	0.500	0.286	0.333	0.400	0.250	0.286	0.333	0.286	0.333	0.400
40	1.200	1.300	1.350	0.400	0.500	0.667	0.333	0.400	0.500	1.100	1.150	1.200

Note: PIPRECIA = PIvot PAirwise RElative Criteria Importance Assessment; C1: Railway line capacity; C2: Railway station capacity; C3: Number of passed trains; C4: Quality of train operations. *l*, *m*, and *u* denote the lower, modal, and upper values of the triangular fuzzy number, respectively.

Before forming the matrix of relative weights of the criterion $\bar{w}'_j$, as an intermediate step, the values for $\bar{q}'_j$ are summed up and the values obtained are: (3.350; 3.990; 4.561).

By applying Eq. (9), the matrix of relative weights of the criterion $\bar{w}'_j$ is formed.

$$\bar{w}'_j = \begin{bmatrix} 0.091 & 0.155 & 0.254 \\ 0.098 & 0.150 & 0.225 \\ 0.145 & 0.210 & 0.284 \\ 0.182 & 0.235 & 0.299 \\ 0.219 & 0.251 & 0.299 \end{bmatrix}$$

After applying the equation: $DF = \frac{l+4m+u}{6}$, the defuzzy values obtained for the PIPRECIA-Inverse method are: 0.161; 0.154; 0.211; 0.237 and 0.253.

The full results of the previous calculations are shown in Table 5 and Table 6. The penultimate columns (6) of these tables show the defuzzified values of the relative weights of the criteria.

Table 5. Calculation and weight values of the main criteria according to the PIPRECIA method

PIPRECIA	s_j	k_j	q_j	w_j	Defuzzy	Rank
1	2	3	4	5	6	7
C1		1.000, 1.000, 1.000	1.000, 1.000, 1.000	0.148, 0.184, 0.243	0.188	4
C2	0.709, 0.847, 0.962	1.038, 1.153, 1.291	0.775, 0.867, 0.963	0.115, 0.160, 0.234	0.164	5
C3	1.112, 1.275, 1.327	0.673, 0.725, 0.888	0.872, 1.197, 1.430	0.129, 0.220, 0.347	0.226	3
C4	0.898, 1.012, 1.122	0.878, 0.988, 1.102	0.792, 1.211, 1.629	0.117, 0.223, 0.395	0.234	1
C5	0.843, 0.952, 1.059	0.941, 1.048, 1.157	0.684, 1.156, 1.731	0.101, 0.213, 0.420	0.229	2
Total			4.123, 5.431, 6.754		1.041	

Note: PIPRECIA = Pivot Pairwise Relative Criteria Importance Assessment; C1: Railway line capacity; C2: Railway station capacity; C3: Number of passed trains; C4: Quality of train operations; C5: Revenues of the planned timetable.

Table 6. Calculation and values of the weights of the main criteria according to the PIPRECIA-Inverse method

PIPRECIA-I	s_j^*	k_j^*	q_j^*	w_j^*	Defuzzy-I	Rank-I
1	2	3	4	5	6	7
C1	0.919, 1.035, 1.115	0.885, 0.965, 1.081	0.414, 0.620, 0.852	0.091, 0.155, 0.254	0.161	4
C2	0.525, 0.601, 0.738	1.262, 1.399, 1.475	0.447, 0.598, 0.754	0.098, 0.150, 0.225	0.154	5
C3	0.744, 0.881, 0.947	1.053, 1.119, 1.256	0.660, 0.836, 0.952	0.145, 0.210, 0.284	0.211	3
C4	0.794, 0.932, 1.003	0.997, 1.068, 1.206	0.829, 0.936, 1.003	0.182, 0.235, 0.299	0.237	2
C5		1.000, 1.000, 1.000	1.000, 1.000, 1.000	0.219, 0.251, 0.299	0.253	1
Total			3.350, 3.990, 4.561		1.016	

Note: PIPRECIA = Pivot Pairwise Relative Criteria Importance Assessment; C1: Railway line capacity; C2: Railway station capacity; C3: Number of passed trains; C4: Quality of train operations; C5: Revenues of the planned timetable.

The Pearson correlation coefficient [1] for the obtained ranking is 0.963, indicating that the values of the criteria within fuzzy PIPRECIA and its inverse steps are almost fully correlated. The Spearman correlation coefficient for the obtained ranking is 0.900, indicating that these rankings are highly correlated. Finally, the obtained relative weights of the criteria, $\bar{w}_j$ and $\bar{w}'_j$, are aggregated, and the final weight values $\bar{w}''_j$ obtained for the main criteria are: 0.174; 0.159; 0.219; 0.235 and 0.241, which is also shown in Table 7.

Table 7. Weight values of the main criteria according to the fuzzy PIPRECIA method

Criterion	Defuzzy	Defuzzy-I	w_j^*	Rank
C1	0.188	0.161	0.174	4
C2	0.164	0.154	0.159	5
C3	0.226	0.211	0.219	3
C4	0.234	0.237	0.235	2
C5	0.229	0.253	0.241	1

Note: PIPRECIA = Pivot Pairwise Relative Criteria Importance Assessment; C1: Railway line capacity; C2: Railway station capacity; C3: Number of passed trains; C4: Quality of train operations; C5: Revenues of the planned timetable.

The final significance order of the criteria is as follows: C5: Revenues of the planned timetable, C4: Quality of train operations, C3: Number of passed trains, C1: Railway line capacity, C2: Railway station capacity.

To assess the robustness of the obtained ranking, a one-at-a-time sensitivity analysis was conducted. The weight of each criterion was perturbed by $\pm 10\%$ and $\pm 20\%$, while the weights of the remaining criteria were proportionally rescaled to maintain the unit sum constraint. The results, presented in Table 8, indicate that across all 20 tested

scenarios the five criteria consistently separate into two stable groups: a high-importance group comprising C3, C4 and C5 (always occupying ranks 1–3) and a lower-importance group comprising C1 and C2 (always occupying ranks 4–5). The boundary between the two groups is robust under all perturbations. The relative ordering within each group, however, is sensitive to the perturbations—a finding consistent with the small numerical differences between the final weights of the criteria within each group. Examples of sensitivity analysis can also be found in the studies by Wolters and Mareschal [26] and Pamučar et al. [27].

Table 8. Sensitivity analysis of criteria weights

Scenario	C1	C2	C3	C4	C5	Ranking	Note
Base case	0.174	0.159	0.219	0.235	0.241	C5 > C4 > C3 > C1 > C2	from Table 7
C1 –20%	0.139	0.166	0.228	0.245	0.251	C5 > C4 > C3 > C2 > C1	within low cluster
C1 –10%	0.157	0.162	0.224	0.240	0.246	C5 > C4 > C3 > C2 > C1	within low cluster
C1 +10%	0.191	0.156	0.215	0.230	0.236	C5 > C4 > C3 > C1 > C2	stable
C1 +20%	0.209	0.153	0.210	0.225	0.231	C5 > C4 > C3 > C1 > C2	stable
C2 –20%	0.180	0.127	0.227	0.244	0.250	C5 > C4 > C3 > C1 > C2	stable
C2 –10%	0.177	0.143	0.223	0.239	0.245	C5 > C4 > C3 > C1 > C2	stable
C2 +10%	0.171	0.175	0.215	0.231	0.237	C5 > C4 > C3 > C2 > C1	within low cluster
C2 +20%	0.168	0.191	0.211	0.226	0.232	C5 > C4 > C3 > C2 > C1	within low cluster
C3 –20%	0.183	0.168	0.175	0.248	0.254	C5 > C4 > C1 > C3 > C2	C3 → 4th
C3 –10%	0.179	0.163	0.197	0.241	0.247	C5 > C4 > C3 > C1 > C2	stable
C3 +10%	0.169	0.155	0.241	0.229	0.234	C3 > C5 > C4 > C1 > C2	within high cluster
C3 +20%	0.165	0.150	0.263	0.222	0.228	C3 > C5 > C4 > C1 > C2	within high cluster
C4 –20%	0.184	0.168	0.232	0.188	0.255	C5 > C3 > C4 > C1 > C2	within high cluster
C4 –10%	0.179	0.164	0.226	0.212	0.248	C5 > C3 > C4 > C1 > C2	within high cluster
C4 +10%	0.169	0.154	0.213	0.259	0.234	C4 > C5 > C3 > C1 > C2	within high cluster
C4 +20%	0.164	0.150	0.206	0.282	0.227	C4 > C5 > C3 > C1 > C2	within high cluster
C5 –20%	0.185	0.169	0.232	0.249	0.193	C4 > C3 > C5 > C1 > C2	within high cluster
C5 –10%	0.179	0.164	0.226	0.242	0.217	C4 > C3 > C5 > C1 > C2	within high cluster
C5 +10%	0.169	0.154	0.212	0.228	0.265	C5 > C4 > C3 > C1 > C2	stable
C5 +20%	0.163	0.149	0.206	0.221	0.289	C5 > C4 > C3 > C1 > C2	stable

Note: C1: Railway line capacity; C2: Railway station capacity; C3: Number of passed trains; C4: Quality of train operations; C5: Revenues of the planned timetable. Each scenario perturbs one criterion's weight by the indicated percentage; the remaining four weights are proportionally rescaled so that the sum equals one.

5 Conclusion

This study evaluated the relative importance of the principal criteria influencing railway timetable planning using the fuzzy PIPRECIA method. Five dimensions representing infrastructure capability, operational throughput, service performance, and economic outcomes were assessed through expert evaluation involving forty decision-makers from different professional backgrounds. The obtained results identified revenues of the planned timetable as the most influential criterion, followed by quality of train operations and number of passed trains, while railway line capacity and railway station capacity received comparatively lower weights.

Beyond the overall ranking, the sensitivity analysis provided additional insight into the structure of timetable planning priorities. The results showed that the five criteria formed two relatively stable groups: a high-importance cluster consisting of number of passed trains, quality of train operations, and revenues of the planned timetable, and a lower-importance cluster represented by railway line capacity and railway station capacity. This grouping remained stable under perturbations of up to $\pm 20\%$ of individual criterion weights. At the same time, the internal ordering within each group showed sensitivity to relatively small changes in weight values and therefore should not be interpreted as an absolute hierarchy.

From a transportation system perspective, the findings indicate that timetable planning decisions are influenced more strongly by operational and economic performance than by infrastructure-capacity indicators alone. Infrastructure-related dimensions remain essential because they establish the operating conditions under which transport services are delivered, but their direct contribution to timetable prioritisation appears lower within the evaluated decision context. These findings support the interpretation of railway timetable planning as a coordinated operational decision process rather than solely an infrastructure allocation problem.

Several limitations should be acknowledged. First, the expert panel was primarily composed of infrastructure managers and railway operators within a single national setting, which may have affected the relative importance assigned to economic criteria. Second, the five criteria were evaluated as independent dimensions although operational interactions exist among them. Railway line capacity and station capacity directly influence transport

throughput, which subsequently affects operational quality and revenue generation. Third, the twenty-five sub-criteria introduced in Section 2 were not independently weighted. Finally, the obtained criterion weights were not applied to the evaluation of alternative timetable scenarios.

Future research may extend the present work in several directions. The hierarchical relationships between criteria and sub-criteria may be examined jointly through extensions capable of capturing criterion interdependence. The obtained weights may also be incorporated into integrated decision-support procedures, including combinations with ranking approaches such as technique for order preference by similarity to ideal solution, MABAC, or evaluation based on distance from average solution for evaluating alternative timetable schemes. In addition, broader international expert participation and subgroup comparison across managerial, engineering, and academic perspectives may provide further evidence regarding the stability and transferability of the identified planning priorities.

Overall, this study provides a structured basis for understanding the relative importance of key planning dimensions and contributes a decision-support perspective for improving railway timetable planning and transportation system operations.

Author Contributions

Conceptualization, M.V., S.V., and S.M.; methodology, M.V. and Ž.S.; validation, S.V. and S.M.; investigation, M.V.; data curation, M.V.; writing—original draft preparation, M.V. and Ž.S.; writing—review and editing, S.V. and S.M.; supervision, S.V. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data supporting our research results are included within the article or supplementary material.

Conflicts of Interest

The authors declare no conflicts of interest.

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